



CHILD MONITORING Circuit

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This small circuit, in two parts, helps in ensuring that a child does not get lost in a crowd during big festivals. It includes

a transmitter and a receiver. The transmitter is attached to the small child and the receiver remains with the parents or the guardian accom-

panying the child. During operation, if the child moves away beyond the set range, the receiver starts vibrating to alert the parents.

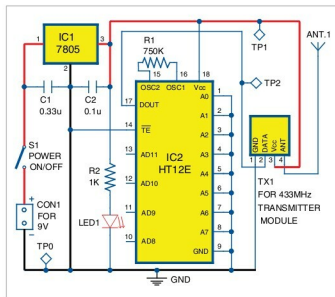


Fig. 1: Transmitter circuit

PARTS LIST

Semiconductors:

- IC1, IC4 - 7805, 5V voltage regulator
- IC2 - HT12E encoder
- IC3 - HT12D decoder
- T1 - BC558 pnp transistor
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 750-k Ω -ohm
- R2 - 1-k Ω -ohm
- R3 - 33-k Ω -ohm
- R4 - 10-k Ω -ohm

Capacitors:

- C1, C3 - 0.33 μ F ceramic disk
- C2, C4 - 0.1 μ F ceramic disk

Miscellaneous:

- TX1 - 433MHz transmitter module
- RX1 - 433MHz receiver module
- CON1, CON2 - 2-pin terminal connector
- CON3, CON4 - 2-pin connector
- S1, S2 - On/off switch
- ANT.1, ANT.2 - 10-17cm long wire
- 9V regulated power supply or 9V battery
- ERM vibrator (3V)
- 3V battery/power supply

Circuit and working

Fig. 1 shows the transmitter circuit and Fig. 2 the receiver circuit. The system uses a set of 433MHz transmitter and receiver modules, encoder modules, and decoder (HT12E) and decoder (HT12D) ICs, 7805 voltage regulator, BC558 transistor and an eccentric rotating mass (ERM) vibrator. The 18-pin

encoder (IC2) in transmitter encodes the 4-channel transmitter signal. It can be addressed by using the active-low address pins.

The decoder (IC3), also an 18-pin IC, used in receiver decodes the transmitter's signal into 4-channel signals. It has the same address pins as the encoder. IC3 produces active-low signals. Pin 17 of decoder IC3 is used to indicate when the connection gets established between the transmitter and the receiver.

RF modules' terminals are clearly marked. Both the modules are connected to 5V supplies. Each module uses

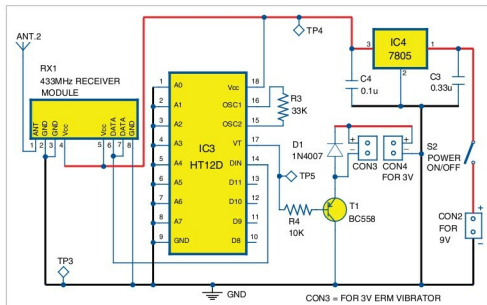


Fig. 2: Receiver circuit